

Carryover-sensitivity study: development-run review

96-cell reduced grid, 150 replicates per cell (2026-05-20)

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1 Overview

This document presents pipeline-verification results for the carryover-sensitivity manuscript from the development simulation run completed 2026-05-20. The run uses a 96-cell reduced Tier 1 grid (150 replicates per cell, MC SE ≈ 0.041 at $p = 0.5$) and a 31-cell Tier 2 sensitivity grid (150 replicates per cell). All figures and tables are computed inline from the development RDS files; nothing is loaded from pre-rendered figures or the shared `analysis/data/` summaries.

The purpose of this document is to verify that:

- a) the $A2 > A1 \approx A3$ power ranking is visible at 150 reps (direction check, not precision check);
- b) type-I error is below nominal across specifications;
- c) sensitivity-block trends are plausible.

Precise point estimates should be drawn from the production run (500 reps, 540 cells) rather than this document.

Dev grid coverage. The Tier 1 dev grid retains both DGP architectures and all three DGP decay forms (the primary axes), trimming secondary axes to two representative values each. `t1half` is

Table 1: Development-run metadata read from the output RDS files.

Item	Value
Tier 1 cells	96
Tier 1 reps/cell	150
Tier 2 cells	31
Tier 2 reps/cell	150
Total Tier 1 rows	43200
Total Tier 2 rows	13950
Run date	2026-05-20

Table 2: Reference-cell performance (Architecture B, exponential DGP, Hybrid, $N = 70$, $c_{bm} = 0.45$, $n_{sim} = 150$). MC SEs are approximately twice those of the 500-rep production run; treat rankings, not precise values, as informative here.

t1/2	spec	Power (MC SE)	Bias (MC SE)	Emp SE	Coverage (MC SE)	MSE (MC SE)	Non-conv
0	A1	0.967 (0.015)	+0.043 (0.071)	0.870	0.953 (0.017)	0.759 (0.081)	0.000
0	A2	0.967 (0.015)	+0.043 (0.071)	0.870	0.953 (0.017)	0.759 (0.081)	0.000
0	A3	0.967 (0.015)	+0.044 (0.071)	0.866	0.953 (0.017)	0.752 (0.080)	0.000
1	A1	0.727 (0.036)	+1.231 (0.065)	0.792	0.767 (0.035)	2.143 (0.165)	0.000
1	A2	0.873 (0.027)	+0.098 (0.081)	0.997	1.000 (0.000)	1.003 (0.105)	0.000
1	A3	0.720 (0.037)	+1.249 (0.066)	0.804	0.767 (0.035)	2.206 (0.170)	0.000

reduced to $\{0.0, 1.0\}$ weeks; design to $\{CO, Hybrid\}$; N to $\{35, 70\}$; c_{bm} to $\{0.0, 0.45\}$; Weibull uses only $k = 0.7$. The OLBDC design and $k = 1.5$ Weibull shape are absent from the dev grid; any figure filtered to those levels will appear empty.

2 Tier 1 results

2.1 Headline performance at the reference cell

Table 2 shows the Morris-style performance measures at the reference cell (Architecture B, exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$, both carryover half-lives present in the dev grid).

2.2 Power by analysis specification

2.3 Decay-form by analysis-spec heatmap

2.4 Type-I error control

2.5 Summary table: all cells at $N = 70$, $c_{bm} = 0.45$, Hybrid

Table 3: Power and non-convergence at $N = 70$, $c_{bm} = 0.45$, Hybrid design, dev grid (150 reps per cell). All unique DGP-architecture combinations present in the dev grid.

Arch	DGP	t1/2	spec	Power (MC SE)	Non-conv
A	exponential	0	A1	0.933 (0.020)	0.000
A	exponential	0	A2	0.933 (0.020)	0.000
A	exponential	0	A3	0.940 (0.019)	0.000
A	exponential	1	A1	0.960 (0.016)	0.000
A	exponential	1	A2	0.920 (0.022)	0.000
A	exponential	1	A3	0.960 (0.016)	0.000
A	linear	0	A1	0.967 (0.015)	0.000
A	linear	0	A2	0.967 (0.015)	0.000
A	linear	0	A3	0.967 (0.015)	0.000
A	linear	1	A1	0.967 (0.015)	0.000
A	linear	1	A2	0.867 (0.028)	0.000
A	linear	1	A3	0.960 (0.016)	0.000
A	weibull k=0.7	0	A1	0.993 (0.007)	0.000
A	weibull k=0.7	0	A2	0.993 (0.007)	0.000
A	weibull k=0.7	0	A3	1.000 (0.000)	0.000
A	weibull k=0.7	1	A1	0.973 (0.013)	0.000
A	weibull k=0.7	1	A2	0.947 (0.018)	0.000
A	weibull k=0.7	1	A3	0.973 (0.013)	0.000
B	exponential	0	A1	0.967 (0.015)	0.000
B	exponential	0	A2	0.967 (0.015)	0.000
B	exponential	0	A3	0.967 (0.015)	0.000
B	exponential	1	A1	0.727 (0.036)	0.000
B	exponential	1	A2	0.873 (0.027)	0.000
B	exponential	1	A3	0.720 (0.037)	0.000
B	linear	0	A1	0.980 (0.011)	0.000
B	linear	0	A2	0.980 (0.011)	0.000
B	linear	0	A3	0.980 (0.011)	0.000
B	linear	1	A1	0.813 (0.032)	0.000
B	linear	1	A2	0.927 (0.021)	0.000
B	linear	1	A3	0.813 (0.032)	0.000
B	weibull k=0.7	0	A1	1.000 (0.000)	0.000
B	weibull k=0.7	0	A2	1.000 (0.000)	0.000
B	weibull k=0.7	0	A3	1.000 (0.000)	0.000
B	weibull k=0.7	1	A1	0.740 (0.036)	0.000
B	weibull k=0.7	1	A2	0.780 (0.034)	0.000
B	weibull k=0.7	1	A3	0.740 (0.036)	0.000

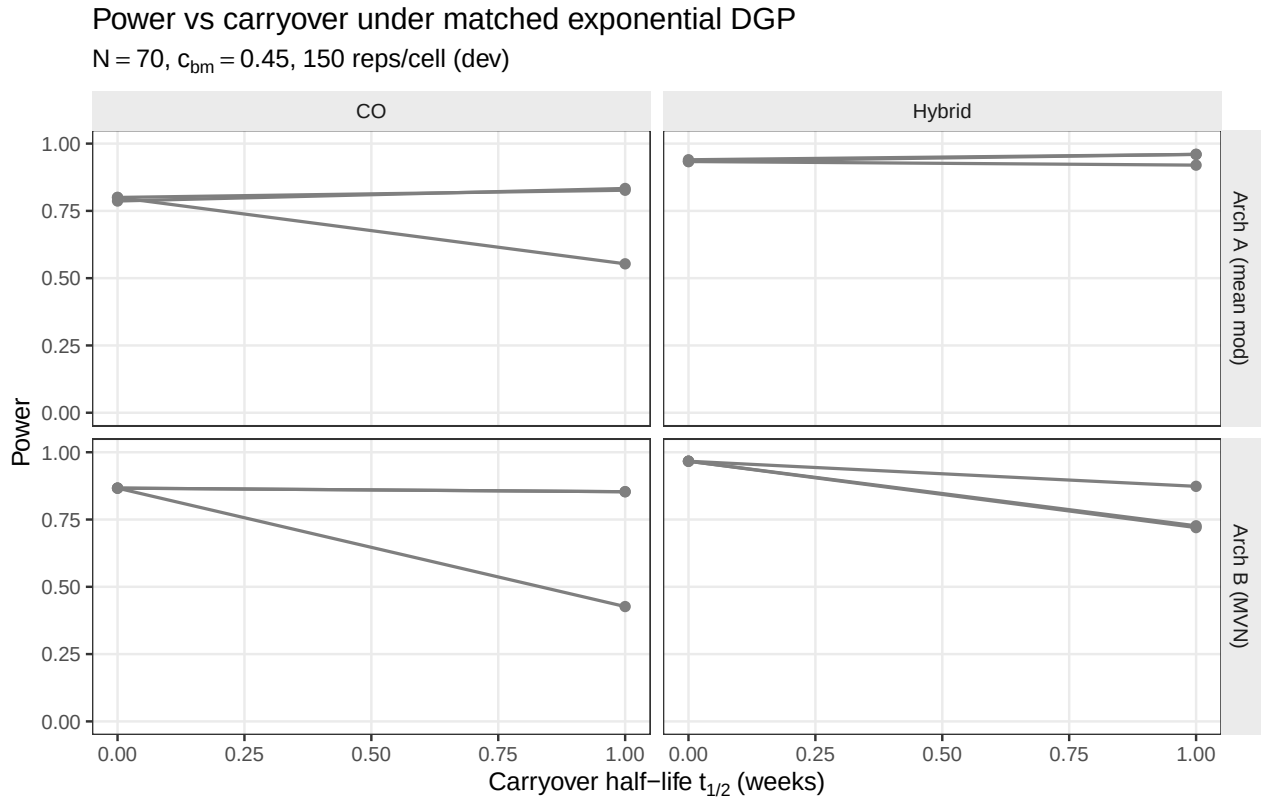


Figure 1: Power versus carryover half-life under the exponential DGP, stratified by architecture and design. Dev grid: design in {CO, Hybrid} only; $N = 70$, $c_{bm} = 0.45$, 150 reps per cell. MC SE ≈ 0.041 at $p = 0.5$.

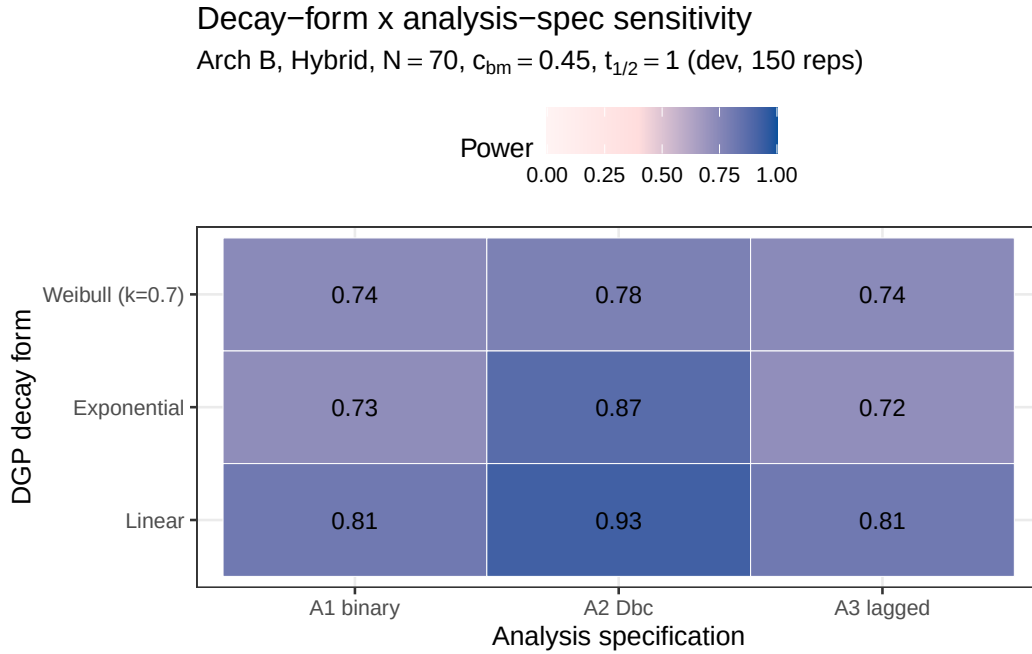


Figure 2: Power at the reference carryover cell ($t_{1/2} = 1.0$ week, Architecture B, Hybrid design, $N = 70$, $c_{bm} = 0.45$) across DGP decay forms and analysis specifications. Dev grid contains three DGP forms: Linear, Exponential, and Weibull $k = 0.7$.

Type-I error control under null

$c_{bm} = 0$, $N = 70$, pooled across DGP forms and carryover levels (dev)

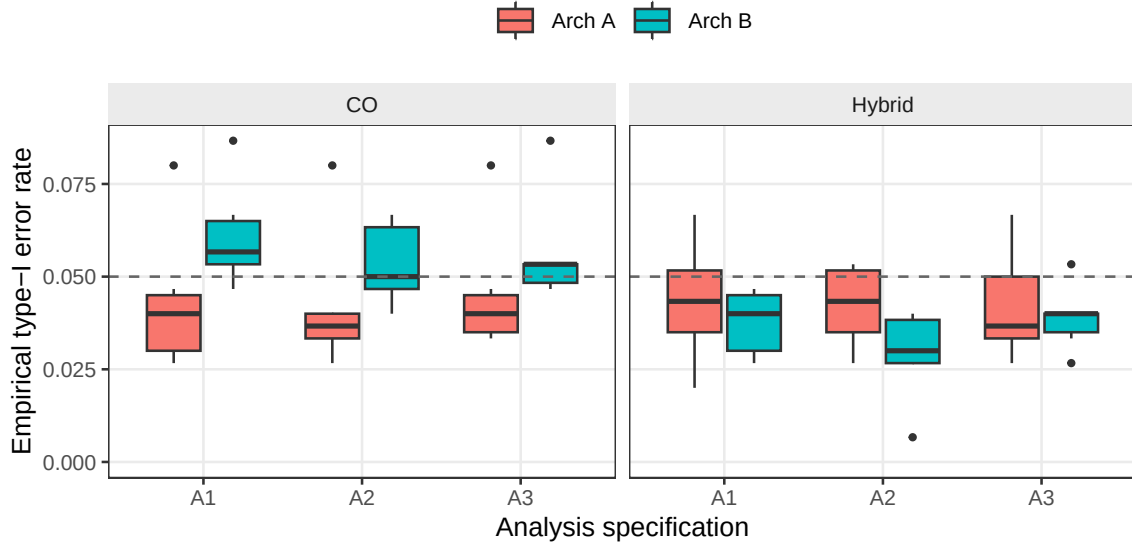


Figure 3: Empirical type-I error rates under the null ($c_{bm} = 0$), pooled across DGP decay forms and carryover half-lives, stratified by design and analysis specification. $N = 70$; dashed line at nominal 0.05. Dev grid includes CO and Hybrid designs only.

S1: Sensitivity to within-factor autocorrelation

Hybrid, Arch B, $N=70$, $c_{bm}=0.45$, exp DGP (dev, 150 reps)

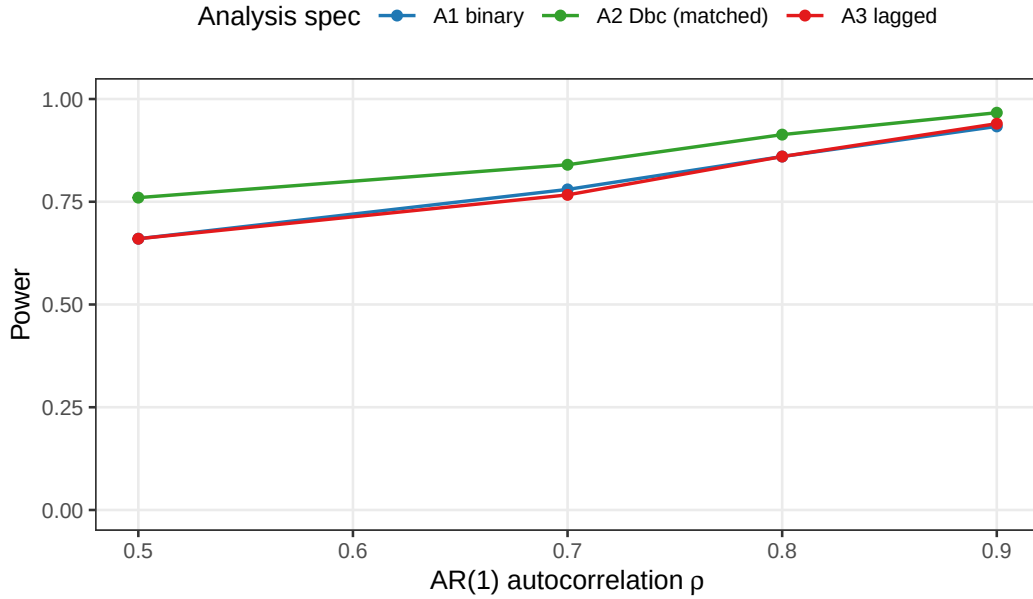


Figure 4: Power by analysis specification across AR(1) autocorrelation $\rho \in \{0.5, 0.7, 0.8, 0.9\}$. Reference: Arch B, Hybrid, $N = 70$, $c_{bm} = 0.45$, exponential DGP, $t_{1/2} = 1.0$ weeks.

S2: Cost of analyst-truth half-life mismatch (dev)

A1 and A3 do not depend on assumed half-life; A2 does

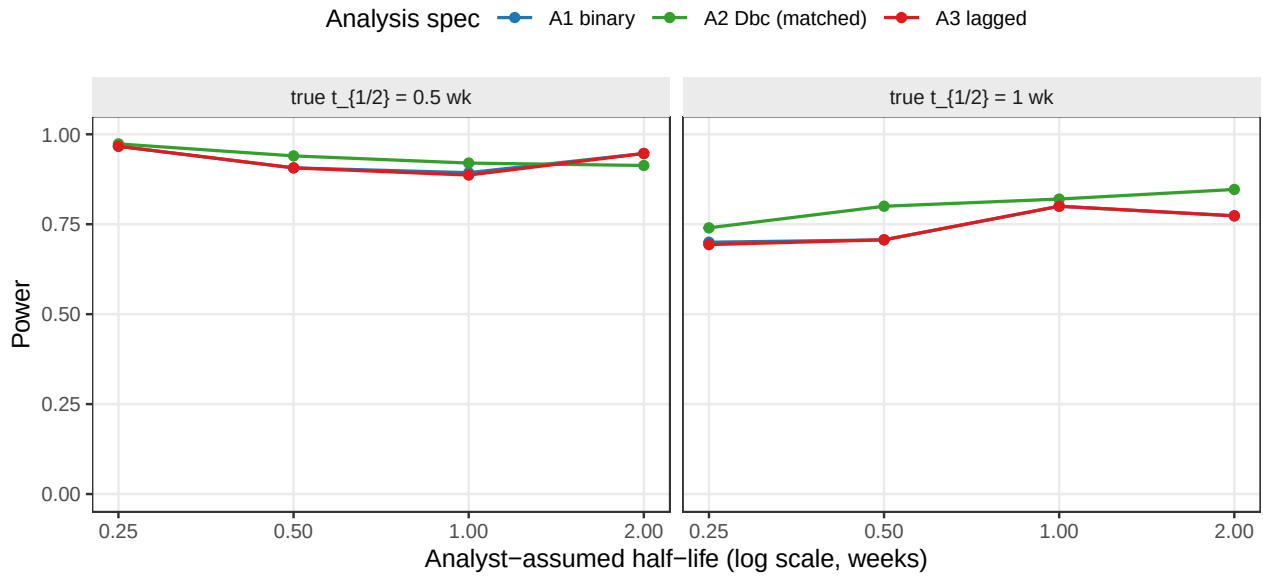


Figure 5: Power of A1, A2, and A3 as the analyst's assumed half-life departs from the true DGP half-life. A1 and A3 are invariant to the assumed half-life by construction. Facets show the two true DGP half-lives present in the dev grid.

S3: Sensitivity to dropout (dev)

MCAR and MAR (severity-biased); reference cell as above

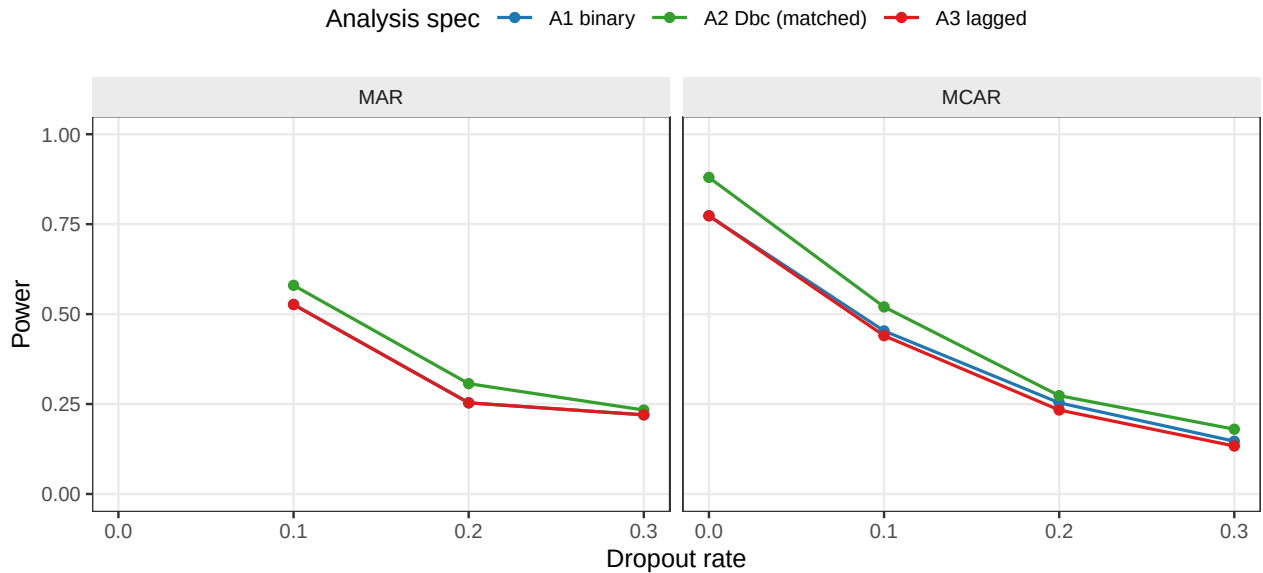


Figure 6: Power at dropout rates $\in \{0, 0.1, 0.2, 0.3\}$ under MCAR and MAR mechanisms.

S4: Biomarker-moderation effect-size curve (dev)

Dashed line at nominal $\alpha = 0.05$

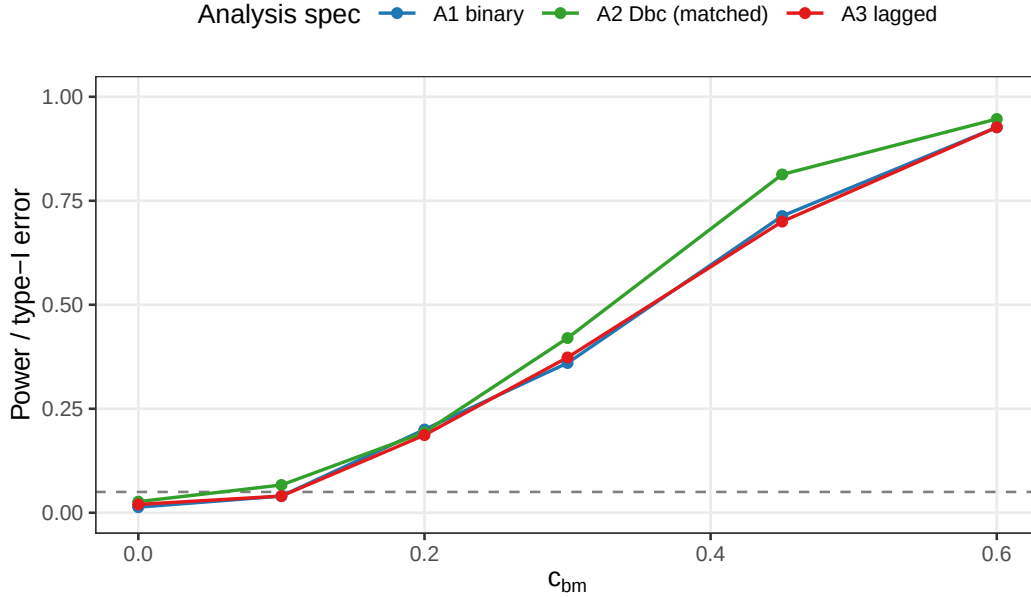


Figure 7: Power across $c_{bm} \in \{0, 0.10, 0.20, 0.30, 0.45, 0.60\}$. The $c_{bm} = 0$ point is the null; dashed line at nominal $\alpha = 0.05$.

Table 4: Automated sanity checks against the dev-run results. Thresholds are deliberately loose (MC SE 0.041 at 150 reps).

Check	Result
A2 power > A1 power at reference cell (Hybrid, Arch B, $t1/2=1$)	A2=0.873, A1=0.727 -> PASS
Max type-I error (N=70) below 0.15	Max=0.087 -> PASS
Max non-convergence rate below 0.20	Max=0.000 -> PASS

3 Tier 2 sensitivity results

3.1 S1: within-factor autocorrelation

3.2 S2: analyst-truth half-life mismatch

3.3 S3: dropout

3.4 S4: biomarker-moderation effect-size curve

3.5 S5: autocorrelation by carryover (exploratory)

4 Dev-run review checklist

Dev review document rendered from *analysis/report/02-carryover-sensitivity/report-devresults.Rmd* on 2026-05-20 at 10:39 PDT. Source data: *output/01-dev.rds* (43200 rows), *output/04-sensitivity-dev.rds* (13950 rows).

S5: Autocorrelation x carryover interaction (dev)

Is the spec ranking rho-sensitive?

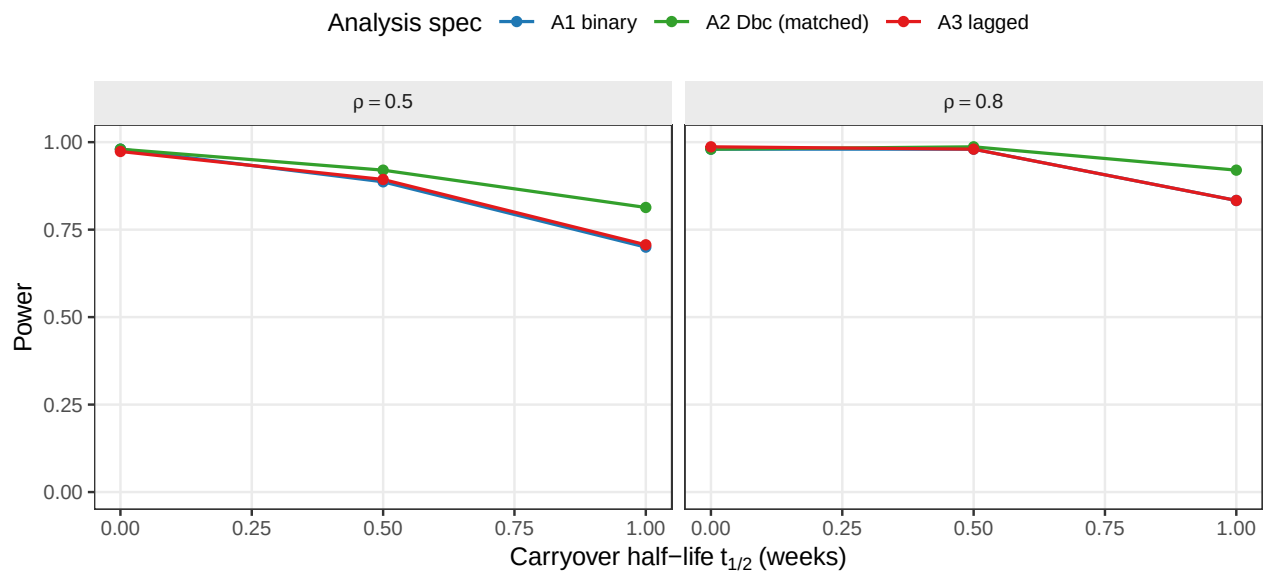


Figure 8: Power across carryover half-lives $\{0, 0.5, 1.0\}$ weeks at $\rho \in \{0.5, 0.8\}$. Facets by ρ .